From Scores to Steps

Diagnosing and Improving LLM Performance in Evidence-Based Medical Calculations

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Presentation Outline

1. Motivation and Research Problem
2. Step-wise Evaluation Framework
3. MedRaC Methodology
4. Experiments and Results
5. Local Demo and Mini-Experiment
6. Discussion and Conclusion

Evidence-Based Medical Calculation as a Structured Task

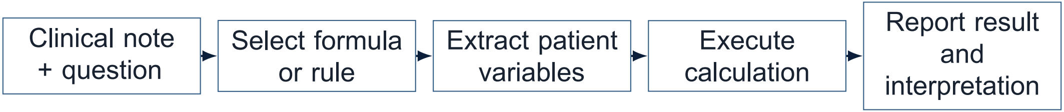

Figure 1. Evidence-based medical calculation requires a sequence of dependent decisions.

Equation-based calculators

$$y = f(x_1, x_2, \ldots, x_n) \qquad (1)$$

Examples: BMI, corrected sodium, and Cockcroft-Gault creatinine clearance.

Rule-based calculators

$$\text{Score} = \sum_{j=1}^{m} p_j \qquad (2)$$

Examples: Wells score and CHA_2DS_2-VASc.

An error at any stage can invalidate the reported clinical result.

[1]

Case Study: A Correct Score from an Incorrect Formula

1. Clinical task

A 57-year-old man has serum sodium 127 mEq/L and glucose 527 mg/dL. Compute corrected sodium using the Hillier equation.

2. Reference calculation

$$\begin{aligned} Na_{\text{corr}} &= Na_{\text{measured}} + 0.024\left(\text{Glucose} - 100\right) \\ &= 127 + 0.024(527 - 100) = 137.248 \end{aligned} \tag{3}$$

3. Model calculation

The model extracts the correct values but applies a different correction rule:

$$127 + 0.016(527) = 135.432 \tag{4}$$

4. Evaluation outcome

Baseline score:

Correct $(135.432 \in [130.39, 144.11])$

Step-wise score: *F* **Incorrect**; *E* **Correct**; *C* **Correct**; *A* **Incorrect**

The numerical tolerance hides a formula-selection error; the acceptable value is not a valid solution.

[1]

Research Gap and Questions

Limitations of prior evaluation

- Final-answer scoring does not verify the selected medical formula.
- Broad numerical tolerances can accept an invalid computation path.
- Aggregate accuracy does not identify the first failed reasoning stage.
- Existing evaluations conflate knowledge, extraction, interpretation, and arithmetic errors.

[1]

Research questions

1. How should each stage of a medical calculation be evaluated?
2. Which stage causes the first failure?
3. Can formula retrieval and code execution reduce these failures?

Main Contributions

1. **Step-wise evaluation.** Evaluate each MedCalc-Bench case at Formula, Extraction, Calculation, and Answer.
2. **Structured diagnosis.** Add Conditional Correctness, First Error Attribution, and an eight-category error taxonomy.
3. **Training-free intervention.** Combine Formula RAG and Python execution in MedRaC; the largest reported gain is 53.19 percentage points.

Evaluate	Diagnose	Intervene
F–E–C–A	Error attribution	Retrieval + execution

For GPT-4o, reported accuracy drops from 62.7% under final-answer scoring to 43.6% under step-wise evaluation.

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Medical Calculation as a Multi-Stage Process

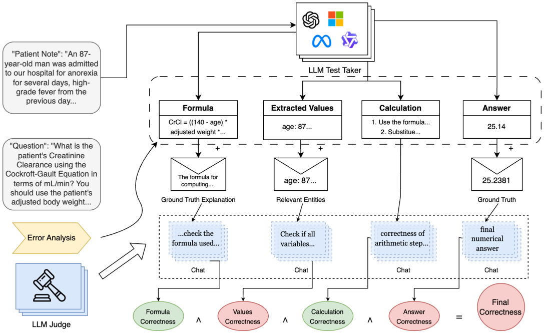

Figure 2. Step-wise evaluation of Formula, Extraction, Calculation, and Answer.

Patient Note + Question → F: Formula → E: Extraction → C: Calculation → A: Answer

Each stage receives a separate correctness judgment, while later stages remain dependent on preceding outputs.

[1]

Formula Selection and Value Extraction

Formula selection, *F*

- Identify the intended calculator or scoring rule.
- Specify its canonical formula, coefficients, units, and boundary conditions.
- Compare the proposed formula with the 55-calculator reference library.

Common failure: formula misselection or hallucination.

[1]

Value extraction, *E*

- Extract every required numerical and categorical variable.
- Preserve units, demographic attributes, and clinical meaning.
- Require complete agreement with the reference annotation.

Common failure: missing, hallucinated, or incorrectly labelled variables.

Calculation Correctness Is Not Answer Correctness

Calculation, *C*

Validity of arithmetic operations given the selected formula and extracted values.

Final answer, *A*

Agreement with the reference value under the paper's precision-aware rule, including valid unit conversions.

$$V(C) = 1, \quad V(F) = 0, \quad V(A) = 0 \tag{5}$$

Arithmetic may be internally consistent while implementing the wrong medical formula.

Precision-aware tolerance: $10.7 \Rightarrow \epsilon = 0.05$; $10.65 \Rightarrow \epsilon = 0.005$.

Outputs with more than two decimals are rounded to two decimals before evaluation.

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Strict Overall Correctness

$$\kappa = V(F) \land V(E) \land V(C) \land V(A) \tag{6}$$

$V(F)$	$V(E)$	$V(C)$	$V(A)$	κ
True	True	True	True	**Correct**
True	True	False	True	**Incorrect**
False	True	True	True	**Incorrect**

Table 1. Strict case correctness under the F–E–C–A decomposition.

[1]

Conditional Correctness

$$CC_i = P(V(S_i) \mid V(S_1) \wedge \cdots \wedge V(S_{i-1})) \tag{7}$$

Interpretation
Reliability of stage S_i after all preceding stages have succeeded.

Example
Calculation CC measures $P(V(C) \mid V(F) \wedge V(E))$, not the unconditional calculation accuracy.

[1]

First Error Attribution

$$FE_i = P(V(S_1) \wedge \cdots \wedge V(S_{i-1}) \wedge \neg V(S_i) \mid \neg \kappa) \tag{8}$$

F $\rightarrow$ **E** $\rightarrow$ **C** $\rightarrow$ **A**

Among strictly failed cases, each case is attributed to the earliest incorrect stage.

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Structured Error Taxonomy

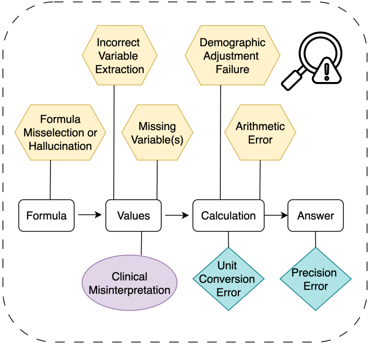

Figure 3. Eight error categories mapped to the calculation pipeline.

Knowledge and formula

Formula Misselection or Hallucination

Extraction and interpretation

Incorrect Variable Extraction; Clinical Misinterpretation; Missing Variable(s); Demographic Adjustment Failure

Numerical execution

Unit Conversion; Arithmetic; Rounding / Precision

Multi-label error agreement is measured with Jaccard similarity: $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$.

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Equation-Based and Rule-Based Tasks

Equation-based

$$y = f(x_1, x_2, \ldots, x_n) \tag{9}$$

- Uses an explicit mathematical transformation.
- Depends on units, coefficients, arithmetic, and rounding.
- Supports deterministic execution directly.

Rule-based

$$\text{Score} = \sum_{j=1}^{m} p_j \tag{10}$$

- Maps clinical criteria to points or categories.
- Requires threshold, presence, and interpretation decisions.
- Is only partly addressed by numerical tools.

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MedRaC Overview

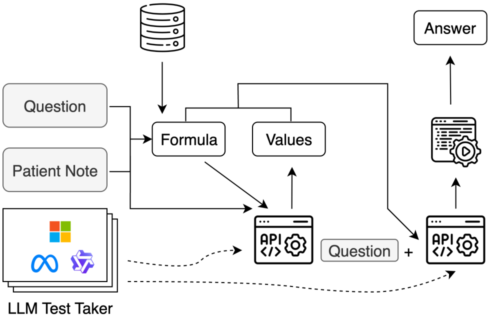

Figure 4. MedRaC combines Formula RAG, value extraction, code generation, and execution.

Essential components

Formula RAG

Value extraction

Code generation

Python execution

Training-free and tool-augmented.

MedRaC is a training-free inference pipeline; it does not introduce a new neural architecture.

[1]

Formula Retrieval

Retrieval procedure

1. Encode the calculator question as e_q.
2. Retrieve d^*, the nearest formula document in D.
3. Ground value extraction and code generation in d^*.

Nearest-neighbor objective

$$d^* = \arg\max_{d \in D} \mathrm{sim}(e_q, e_d) \tag{11}$$

e_q : question embedding
e_d : formula-document embedding

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Value Extraction and Code Generation

$$\underbrace{\text{Patient Note}}_{\text{clinical evidence}} + \underbrace{\text{Question}}_{\text{calculator intent}} + \underbrace{\text{Retrieved Formula}}_{\text{grounded method}} \rightarrow \text{Structured Values} \tag{12}$$

$$\text{Retrieved Formula} + \text{Structured Values} + \text{Question} \rightarrow \text{Python Code} \tag{13}$$

The Patient Note supplies clinical evidence; the Question identifies the requested calculator and output.

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Python-Assisted Calculation

LLM responsibility

- Interpret the clinical text.
- Identify the required patient values.
- Translate the grounded formula into code.

Execution responsibility

```
measured_na = 127
glucose = 527
result = measured_na + 0.024 * (glucose - 100)
```

Python deterministically executes the generated calculation and returns the numerical result.

[1]

How MedRaC Targets Errors

Component	Primary error addressed
Formula RAG	Formula selection and hallucination
LLM value extraction	Required patient variables; remains interpretation-dependent
Code generation	Formula-to-program translation
Python execution	Arithmetic and operation-order errors

Table 2. MedRaC components and their primary target errors.

MedRaC externalizes formula knowledge and arithmetic. Extraction and clinical interpretation remain LLM-dependent.

[1]

Dataset and Benchmark Cleanup

1,048 cases → **108 removed** → **940 retained**

Benchmark scope

- 55 medical calculators sourced from MDCalc.
- Equation-based and rule-based cases.
- 940 cases retained after benchmark curation.

Representative exclusions

- Mistyped gestational-age equation.
- Incorrect APACHE II threshold.
- Reversed limits for negative answers.
- QTc unit mismatch and invalid Caprini scoring.

[1]

Experimental Setup

Cohort	940 retained MedCalc-Bench cases; rule-based and calculation-based tasks
Models	Phi-4-mini; LLaMA3.2-3B; Qwen3-4B/8B/14B; LLaMA3.1-8B; GPT-4o-mini; GPT-4o
Methods	Direct; Zero-shot CoT; One-shot; Self-Refine (≤ 5 rounds); MedPrompt ($k = 3$); MedRaC
Step-wise judge	DeepSeek-chat
Error-type judge	DeepSeek-reasoner
Inference	No training; GPT defaults; open-source models: temperature 0.6, top-p 0.95, repetition penalty 1.0
Hardware	Two NVIDIA RTX A6000 GPUs for local open-source inference

Table 3. Paper experimental configuration.

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Main Results

Model	Direct		CoT		One-shot		Self-Refine		Medprompt		MedRaC	
	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc
Phi-4-mini	16.52	2.16	5.01	2.16	12.09	16.47	2.06	6.16	3.24	10.32	**35.10**	**68.39**
LLaMA3.2-3B	12.39	0.83	1.47	3.99	**14.75**	**18.47**	0.88	2.83	2.95	16.31	-[2]	-
Qwen3-4B	23.30	26.79	9.73	25.79	**49.85**	59.57	9.44	26.29	10.32	45.92	45.72	**68.72**
Qwen3-8B	29.20	42.26	16.52	38.10	**58.70**	62.90	19.76	40.10	15.93	53.74	46.61	**74.54**
LLaMA3.1-8B	19.17	2.50	6.78	8.32	25.07	20.97	5.90	6.82	11.21	28.45	**31.27**	**70.22**
Qwen3-14B	39.53	46.59	26.55	43.43	**60.77**	67.05	27.14	47.25	8.55	22.30	50.44	**78.37**
GPT-4o-mini	22.42	7.99	23.89	34.61	**52.80**	49.58	26.55	33.94	11.80	42.43	50.44	**72.71**
GPT-4o	24.48	13.98	43.07	43.93	**62.24**	54.24	44.84	44.93	25.96	56.91	51.03	**64.39**

Table 4. Paper-reported accuracy across models and prompting methods (%).

Calculation-based

MedRaC has the highest reported Calc accuracy for all seven models with complete results.

Rule-based

Gains vary by model; One-shot is higher for Qwen3-14B and GPT-4o.

Metric note. Direct uses final-answer scoring. Reasoning-based methods use step-wise automatic evaluation.

[1]

Equation-Based and Rule-Based Performance

Model	Rule-based		Calculation-based	
	One-shot	MedRaC	One-shot	MedRaC
Phi-4-mini	12.09	**35.10**	16.47	**68.39**
LLaMA3.1-8B	25.07	**31.27**	20.97	**70.22**
Qwen3-14B	**60.77**	50.44	67.05	**78.37**
GPT-4o	**62.24**	51.03	54.24	**64.39**

Table 5. Selected one-shot and MedRaC accuracy values from the main results (%).

Equation-based

Deterministic execution directly addresses dominant numerical errors.

Rule-based

One-shot examples may better demonstrate how clinical findings map to scoring criteria.

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Error Reduction

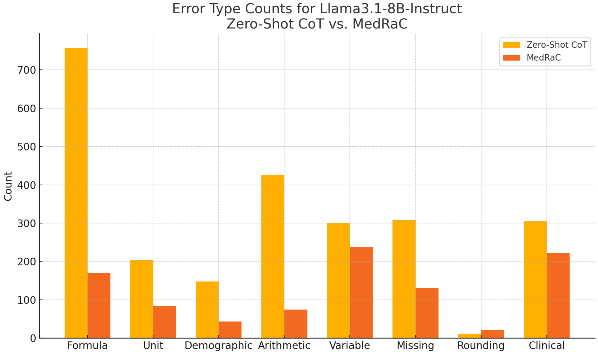


Figure 5. Error counts for LLaMA3.1-8B-Instruct with and without MedRaC.

MedRaC reductions

Formula

$-587\ (-77.5\%)$

Arithmetic

$-352\ (-82.6\%)$

Demographic adjustment

$-105\ (-70.9\%)$

Residual bottlenecks: incorrect extraction falls by 21.3%; clinical misinterpretation falls by 26.9%. Both reductions are smaller than those for formula and arithmetic errors.

[1]

Ablations Isolate the Roles of Retrieval and Execution

Remove Formula RAG

Components	MedRaC	MedRaC w/o RAG
Acc % ↑	64.68	25.64
Formula FE % ↓	20.78	71.96
Formula CC % ↑	92.66	46.49

Table 6. Ablation without Formula RAG.

Accuracy: $64.68\% \to 25.64\%$ (-39.04 pp)

Formula FE: $20.78\% \to 71.96\%$

Remove Code Execution

Components	MedRaC	MedRaC w/o Code
Acc % ↑	64.68	53.09
Calc FE % ↓	3.23	31.88
Calc CC % ↑	97.82	76.52

Table 7. Ablation without code execution.

Accuracy: $64.68\% \to 53.09\%$ (-11.59 pp)

Calculation FE: $3.23\% \to 31.88\%$

Interpretation: retrieval primarily protects formula selection; execution primarily protects calculation.

[1]

LLM-as-Judge Validation

Agreement Type	Formula	Extraction	Calculation	Answer
Expert–Expert	84.8%	84.8%	89.1%	95.7%
Expert–Non-Expert	72.3%	78.1%	66.6%	91.3%
LLM–Expert	90.2%	78.3%	88.1%	97.8%
All Pairs (Overall)	77.2%	81.9%	75.7%	92.5%

Table 8. Agreement between expert and LLM step-wise judgments (%).

Validation set

46 samples from five calculators.

LLM–Expert agreement

F/E/C/A: $90.2/78.3/88.1/97.8\%$. Extraction is lowest and only 0.2 pp above Expert–Non-Expert agreement.

Reliability caveat

The paper reports percent agreement, not a chance-corrected coefficient.

[1]

Local Demo Objective and Pipeline

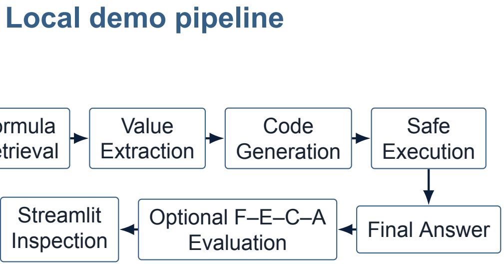


Figure 6. Observable stages in the local seminar pipeline.

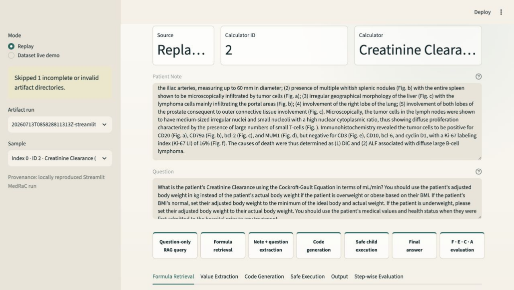
Objective

Execute the workflow and inspect observable intermediate outputs.

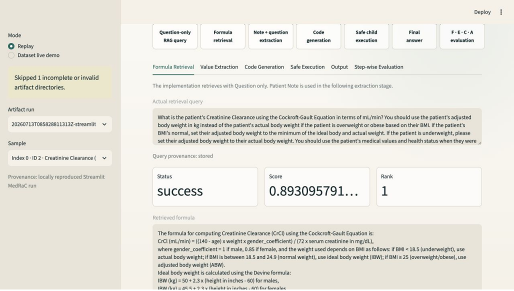
Boundary

The demo uses fixed benchmark rows, exposes structured outputs only, and applies seminar-specific safe execution.

Streamlit Demo and Representative Example



Calculator ID, Patient Note, Question, and observable pipeline stages.



Question-only retrieval, rank, similarity score, and retrieved Cockcroft-Gault formula.

Figure 7. Streamlit replay for Creatinine Clearance (Cockcroft-Gault Equation), Calculator ID 2.

Local Experiment 1: End-Task Evaluation

Method	Correct	Accuracy
Plain CoT	9/10	90%
MedRaC + RAG	**10/10**	100%

Table 9. Rule-based final-answer accuracy on ten fixed cases.

Observation: MedRaC is correct on one additional case ($+10$ percentage points); the result is descriptive for $n = 10$.

Local Experiment 2: Step-wise Evaluation

Method	Formula	Extraction	Calculation	Answer	Strict
Plain CoT	10/10	9/10	10/10	8/10	8/10
MedRaC + RAG	10/10	**10/10**	10/10	**9/10**	**9/10**

Table 10. Post-hoc F–E–C–A and strict correctness on stored outputs.

Observation: MedRaC improves Extraction, Answer, and Strict by one case; Formula and Calculation remain unchanged.

Scope, Limitations, and Evaluation Caveats

Paper scope and evaluation caveats

- Structured, single-turn tasks; curated English notes, not noisy EHR or multilingual data.
- LLM-as-Judge; human validation: 46 cases, percent agreement only.
- Point estimates only; no confidence intervals or significance tests.
- Requires an accurate, current formula bank.
- Public, de-identified data; research only, not clinical deployment.

[1]

Local experiment limitations

- Ten fixed cases; no basis for statistical inference.
- Model substitution through GitHub Models and a separate 55-formula index.
- Post-hoc step-wise evaluation by an LLM Judge, not human annotators.
- One timeout repair and one answer adjudication are documented in the artifacts.
- Not a paper-exact reproduction.

Conclusions

1. Final-answer accuracy is insufficient for evaluating medical calculations.
2. F–E–C–A evaluation, CC, and FE identify conditional reliability and the earliest failed stage.
3. MedRaC reduces formula and arithmetic errors. Extraction and clinical interpretation remain bottlenecks.

End-task scoring should be complemented by domain-grounded process evaluation.

[1]

Questions and Discussion

1. Is the F–E–C–A decomposition sufficient for more complex clinical reasoning workflows?
2. How should an LLM-as-Judge be calibrated before it is used for high-stakes evaluation?
3. How can a formula bank remain versioned, guideline-aware, and clinically auditable?

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Appendix: Complete Main-Results Table

Model	Direct		CoT		One-shot		Self-Refine		Medprompt		MedRaC	
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Table 4. Complete paper-reported accuracy results (paper Table 1; %).

Direct uses final-answer scoring, whereas reasoning-based methods use step-wise automatic evaluation.

[1]

Appendix: Complete Metric Definitions

Strict correctness

$$\kappa = V(F) \wedge V(E) \wedge V(C) \wedge V(A) \tag{14}$$

Conditional Correctness

$$CC_i = P(V(S_i) \mid V(S_1) \wedge \cdots \wedge V(S_{i-1})) \tag{15}$$

First Error Attribution Rate

$$FE_i = P(V(S_1) \wedge \cdots \wedge V(S_{i-1}) \wedge \neg V(S_i) \mid \neg\kappa) \tag{16}$$

[1]

Appendix: Paper Discrepancies

Topic	Conflicting statements	Consequence
Source cohort	Section 2: 1,047 vignettes; Section 4 and Appendix A: 1,048	The reported cleanup total differs by one source case
Formula CC without RAG	Section 5 prose: 7.34%; Table 5: 46.49%	The no-RAG Formula CC value cannot be uniquely reconciled

Table 11. Numerical inconsistencies identified in the paper.

No numerical reconciliation is inferred when the paper does not provide one.

[1]

Appendix: Formula-Bank Scaling

Formula bank: $55 \to 785$ **documents** ($\approx 14\times$)

Embedding model	Top-1 (%)	Top-2 (%)
ada-002	100.00	100.00
text-embedding-3-large	96.36	100.00
text-embedding-3-small	98.18	100.00

Table 12. Retrieval accuracy with the expanded 785-formula memory.

Top-2 retrieval remains 100% in this experiment; the result does not establish scaling to unrestricted medical knowledge.

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Appendix: Local Experiment Provenance

Generation run

Samples	10
Equation / rule	7 / 3
Generator	`gpt-4.1`
Temperature / seed	0.0 / 42
Failed / skipped	0 / 0

Post-hoc evaluation

Evaluator	`gpt-4.1`
Inputs	Stored outputs
Method–sample units	20/20
Judge calls per unit	4
Strict (P / M)	8/10 / 9/10
CC-A (P / M)	88.9 / 90.0%
First errors (P / M)	E+A / A

Table 13. Configuration and selected diagnostic metrics for the local experiment.

Documented artifact amendments

- Stored MedRaC code was re-executed with a 5-second timeout after the original 2-second timeout; no API request was repeated.
- One Answer verdict was adjudicated against the benchmark interval; the original verdict remains in provenance.

Appendix: Safe-Execution Boundary

Implemented safeguards

- AST node and function allowlists.
- No imports, filesystem, network, shell, or input.
- Spawned child process with a strict timeout.
- Result-type and magnitude checks.

Not a secure sandbox

- No kernel isolation.
- No syscall filtering.
- No multi-tenant security guarantee.
- Restricted to the seminar demonstration.

References

B. Wang, I. Xia, Y. Zhang, J. Wang, F. Ouyang, S. Han, A. Cohan, H. Yu, and Z. Yao, "From scores to steps: Diagnosing and improving LLM performance in evidence-based medical calculations," 2026.
arXiv v2, 31 January 2026.